SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

Question 1

Choose the correct answers to the questions from the given options. **[15]**

(Do not copy the questions, write the correct answers only.)

(i) Which gas decolourises potassium permanganate ($KMnO_4$) solution?

(a) Sulphur dioxide

(b) Ammonia

(c) Hydrogen chloride

(d) Carbon dioxide

(ii) Which formula represents a *saturated* hydrocarbon?

(a) C_4H_8

(b) C_5H_{12}

(c) C_4H_6

(d) C_5H_{10}

(iii) The metal whose oxide can be reduced by common reducing agents:

(a) Copper

(b) Sodium

(c) Aluminium

(d) Potassium

(iv) An organic compound has a vapour density of 22. The molecular formula of the organic compound is:

[*Atomic weight: C = 12, H = 1*]

(a) CH_4

(b) C_2H_4

(c) C_2H_6

(d) C_3H_8

(v) In the reaction given below *sulphuric acid* acts as a/an:

$$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$$

(a) Non-volatile acid

(b) Dibasic acid

(c) Oxidising agent

(d) Reducing agent

(vi) **Assertion (A):** The tendency of losing electrons increases down the Group.

Reason (R): The most reactive metal is placed at the top of Group 1.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.

(vii) The ore that can be concentrated by using magnetic separation:

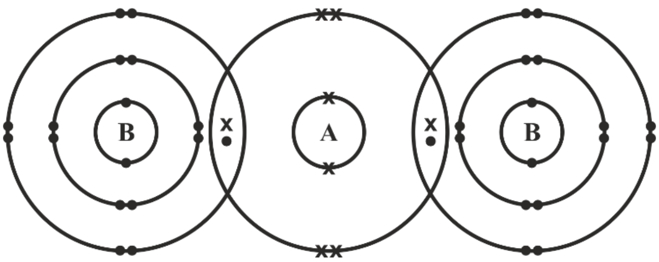
(a) Corundum

(b) Haematite

(c) Calamine

(d) Bauxite

(viii) The diagram given below shows the bonding in the covalent molecule AB_2.



Which option represents the correct electronic configuration of atoms **A** and **B** **before** combining together to form the above molecule?

	A	B
(a)	2, 4	2, 8, 6
(b)	2, 4	2, 8, 7
(c)	2, 8	2, 8, 8
(d)	2, 6	2, 8, 7

(ix) Which of the following options has all the compounds which are members of the *same* homologous series?

(a) CH_4, C_2H_6, C_3H_8

(b) CH_4, C_2H_6, C_3H_6

(c) C_3H_4, C_3H_6, C_3H_8

(d) C_2H_4, C_3H_6, C_4H_{10}

(x) **Assertion (A):** In the *Contact Process* SO_3 gas is not directly dissolved in water to obtain sulphuric acid.

Reason (R): Dense fog or misty droplets of sulphuric acid are formed which is difficult to condense.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.

(xi) Given below are four ions:

Cl^-, Li^+, Al^{3+}, K^+

Identify the pair of ions which have the same electronic configuration.

[*Atomic number: Cl = 17, Li = 3, Al = 13, K = 19*]

(a) Cl^- & Li^+

(b) Al^{3+} & K^+

(c) Cl^- & K^+

(d) Li^+ & K^+

(xii) Which pair of reactants can be **best** used to produce lead (II) sulphate?

(a) Sulphuric acid + Lead

(b) Sulphuric acid + Lead hydroxide

(c) Sodium sulphate + Lead nitrate

(d) Potassium sulphate + Lead oxide

(xiii) Aqueous copper (II) sulphate is electrolysed using copper electrodes.

Which statement about the electrolysis is **not** correct?

(a) An oxidation reaction occurs at the positive electrode.

(b) The current is carried through the electrolyte by ions.

(c) The positive electrode loses mass.

(d) The number of copper (II) ions in the electrolyte decreases.

(xiv) X, Y & Z are three metallic atoms in successive order belonging to the same group such that atomic radii of 'X' is the smallest. Which of the three atoms is the **best** reducing agent?

(a) X

(b) Y

(c) Z

(d) All three have the same reducing power.

(xv) 40 cm^3 of methane (CH_4) is reacted with 60 cm^3 of oxygen.
The equation for the reaction is given below:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

All volumes are measured at room temperature.
What is the **total** volume of the gases remaining at the end of the reaction?

(a) 60 cm^3

(b) 40 cm^3

(c) 45 cm^3

(d) 50 cm^3

MARKING SCHEME

(i)	(a) Sulphur dioxide/SO_2
(ii)	(b) C_5H_{12} / pentane/ correct structural formula as well as condensed form
(iii)	(a) Copper/ Cu
(iv)	(d) C_3H_8 / propane/ correct structural formula as well as condensed form
(v)	(c) Oxidising agent/ Oxidizing/ Oxidizing property/Oxidised/Oxidation/Ox
(vi)	(c) (A) is true but (R) is false.
(vii)	(b) Haematite/ Fe_2O_3/ Ferric oxide/ Iron(III) oxide
(viii)	(d) 2,6 2,8,7
(ix)	(a) CH_4, C_2H_6, C_3H_8 / methane, ethane, propane.
(x)	(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(xi)	(c) Cl^-& K^+ /Chloride ion & Potassium ion/ Chloride & Potassium ion
(xii)	(c) Sodium sulphate + Lead nitrate/ Na_2SO_4 and Pb $(NO_3)_2$
(xiii)	(d) The number of copper (II) ions in the electrolyte decreases.
(xiv)	(c) Z
(xv)	(b) 40 cm^3 /40/ 10 cc CH_4 and 30 cc CO_2

Comments of Examiners

(i) Majority of the candidates answered this question correctly as 'Sulphur dioxide', except for a few candidates who were confused between 'sulphur dioxide' and 'carbon dioxide'.

(ii) Most candidates correctly identified 'C_5H_{12}' as a saturated hydrocarbon. However, some candidates were confused between 'C_5H_{12}' and 'C_5H_{10}' and mistakenly selected 'C_5H_{10}' as a saturated hydrocarbon.

(iii) Many candidates correctly identified 'copper' as the correct option. However, some candidates struggled with the concept of agents 'metallic oxides' and 'reducing agents', leading them to incorrectly choose 'sodium' or 'aluminium'. A few candidates even selected multiple answers.

(iv) Most candidates correctly answered it as 'C_3H_8', though a few chose the incorrect option 'C_2H_6' as they were not clear with the formula.

(v) Most candidates were able to write the correct answer as 'oxidising agent'. Some of the candidates struggled with the concept of 'oxidising agent' and 'reducing agent', leading them to choose the incorrect options.

(vi) Majority of the candidates correctly identified the answer as option (c) that is (A) is true but (R) is false. However, some candidates found it challenging to correlate the assertion (A) and reason (R) statements and incorrectly chose options (a) or (b).

(vii) Most candidates correctly answered this question. However, some were confused and opted for 'bauxite' instead of the correct option 'haematite'. A few others chose 'corundum' or 'calamine', reflecting a lack of clarity about the specific ores.

(viii) Many candidates were not able to comprehend the meaning of 'before combining together'. Several candidates were able to answer this question correctly. Some of the candidates mistakenly chose option (c) as they considered the electronic configuration of the atoms after combination.

(ix) This question was answered well by many candidates, but quite a few were confused between option (a) and option (b) due to lack of understanding of the concept of 'homologous series'. Some candidates chose the incorrect options as (b) and (c). Many candidates failed to identify the general formulas for 'alkanes', 'alkenes', and 'alkynes.

Suggestions for teachers

- Explain the similarity and difference between carbon dioxide and sulphur dioxide with the help of confirmatory tests.
- Demonstrate reactions such as the decolourisation of $KMnO_4$ by SO_2 in laboratory to create strong visual memory.
- Reiterate that both sulphur dioxide and carbon dioxide give positive test for lime water.
- Incorporate confirmatory tests for gases into teaching, along with repeated revision of chemical tests.
- Ensure students engage in lab work to reinforce theoretical concepts with hands-on experience.
- Emphasise on the general formula of homologous series.
- Provide sufficient practice in identifying, naming, and writing structural formulas for saturated and unsaturated hydrocarbons.
- Clearly highlight the comparison among alkanes, alkenes, and alkynes in tabular form by illustrating examples and formulas.
- Conduct regular quizzes and exercises to reinforce identification and classification of organic compounds.
- Explain the concept of reduction and the role of a reducing agent in a chemical reaction.
- Lay stress on the metal reactivity series and how it relates to the stability of metal oxides and the feasibility of reduction.
- Provide practice on reduction of metal oxides, using equation-based examples.
- Use flowcharts or tables to help students to understand the extraction processes.
- Emphasise that higher reactivity of metal means more energy is required for its extraction.
- Explain the correct usage of formula - Molecular mass = 2 × vapour density.
- Clarify the difference between empirical formula, molecular formula, vapour density, and molecular mass with multiple examples.

(x) Many candidates answered this question correctly as both (A) and (R) are true, and (R) is the correct explanation of (A).
Some of the candidates mistakenly considered that 'sulphur trioxide' (SO_3) can be directly dissolved in water and opted for option (d) while some other misunderstood the meaning of 'dense fog' formed during this reaction.

(xi) Most candidates correctly identified the option (c) that is 'Cl^- and K^+'. However, some candidates were confused between options (c) and (d), leading to a significant number of incorrect responses.
Few common errors observed were:
- Number of electrons gained by an element to form an anion.
- Number of electrons lost by an element to form a cation.
- Confusion between Li and K.
- Charge on ion.
- Electronic configuration after ionisation.

(xii) Several candidates failed to write the correct option. Many candidates did not understand the concept of a double displacement reaction and chose the options incorrectly. Only a few candidates were able to identify the correct answer as 'sodium sulphate + lead nitrate'.

(xiii) Most candidates chose the correct answer for this question. However, many candidates overlooked the word 'not' in the question, leading to incorrect answers. A significant number of candidates got it wrong, as they forgot that the concentration of Cu (II) ions remains unchanged when copper electrodes are used.

(xiv) Many candidates were able to answer this question correctly as option (c), while some candidates related the 'small size' of the metallic atom with 'best reducing agent' and selected the incorrect option (a).

Suggestions for teachers

– Provide sufficient practice on calculation of molecular masses by using vapour density.
– Thoroughly teach the chemical properties of sulphuric acid with examples and its role in different reactions.
– Use frequent review and questioning in class to ensure students can clearly explain and apply these concepts.
– Give regular practice of identification of oxidised and reduced reactants in a chemical reaction.
– Give imaginary elements with specific electronic configuration and ask students to demonstrate the bond formation.
– Explain the definition of a homologous series with examples of different organic compounds.
– Emphasise on the general formulas for each homologous series:
  - Alkanes: C_nH_{2n+2}
  - Alkenes: C_nH_{2n}
  - Alkynes: C_nH_{2n-2}
– Clarify the structural features and functional group consistency within a series.
– Provide rigorous practice of writing molecular and structural formulas for members of each homologous series.
– Give ample practice of assertion -reason type questions.
– Explain explicitly 'Contact process' for manufacturing of sulfuric acid by showing animations or videos for better understanding.
– Teach various methods of concentration of ores using suitable videos to reinforce the concepts.
– Encourage students to understand how atoms form cations and anions by losing or gaining electrons respectively with the help of electronic configurations.
– Clarify the difference between atoms and ions and the process of ion formation.
– Provide regular practice exercises on conversion of neutral atoms to ions and writing their corresponding electronic configurations.
– Make students aware of exceptions and special cases in salt preparation.

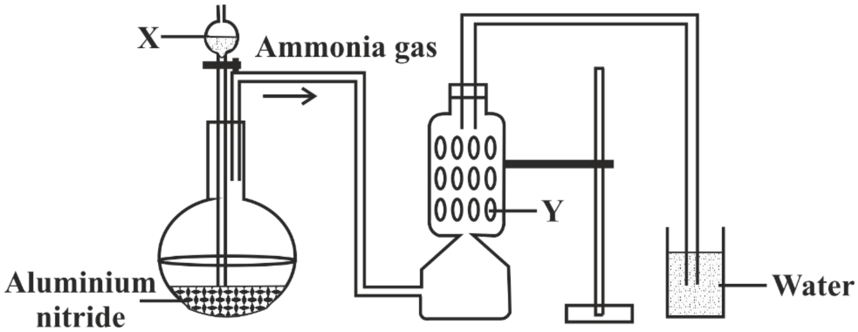
(xv) Majority of the candidates selected the correct answer as '40 cm^3', though many candidates marked '60 cm^3' due to confusion. Some of the candidates calculated the volume of gases formed, but not the total volume of remaining gases at the end of the reaction. A few candidates selected random options.

Suggestions for teachers

- Address specifically why certain methods of salt preparation are preferred over others and explain different methods of salt preparation, including their chemical equations and reaction types
- Conduct simple practical experiments on salt preparation to reinforce the concept through hands-on learning.
- Provide ample practice for writing balanced equations and identifying the type of reaction.
- Teach solubility rules thoroughly, highlighting how they determine the formation of precipitates.
- Stress on the importance of reading the question carefully giving special attention to the word written in bold.
- Explain the basic concept of electrolysis, including the role of electrodes and ion movement.
- Provide detailed observations for both copper (active) electrodes and platinum (inert) electrodes in electrolysis of copper sulphate.
- Emphasise on the difference between electrolysis in terms of equations and observation when carried out using an active electrodes and inert electrodes.
- Highlight how electrode material affects reactions and mass changes.
- Stress the important events during electrolysis, such as ion discharge at electrodes and changes in solution or electrodes.
- Use suitable videos/animations to help students visualise reactions occurring at anode and cathode.
- Provide students within depth knowledge on periodic properties and their variations across period and down the group.
- Provide plenty of practice questions focused on identifying and explaining the periodic trends.
- Link the concept of reducing agent with metallic character of an element.
- Clearly state that Gay Lussac's Law is applicable to gases only.
- Guide the students to pay attention to the physical state of reactants and products while practicing numerical based on Gay Lussac's Law.

Question 2

(i) A student was instructed by the teacher to prepare and collect ammonia gas in the laboratory by using aluminium nitride. The student had set up the apparatus as shown in the diagram below. Study the given diagram and answer the following questions: [5]



(a) Name the substance **X** added through the thistle funnel by the student.

(b) Write a balanced equation for the reaction occurring between Aluminium nitride and substance **X**.

(c) Identify the substance **Y**.

(d) State the function of **Y**.

(e) Why could the student **not collect** ammonia gas at the end of the experiment?

(ii) State the **terms** for the following: [5]

(a) *Undistilled* alcohol containing a large amount of methanol.

(b) A *salt* formed by the *partial* replacement of the *hydroxyl group* of a di-acidic or a tri-acidic base by an acid radical.

(c) Organic compounds having the *same* molecular formula but *different* structural formula.

(d) The tendency of an atom to attract the shared pair of electrons towards itself when combined in a compound.

(e) The type of covalent bond in which electrons are shared *unequally* between the combining atoms.

(iii) Complete the following sentences by choosing the *correct word(s)* from the brackets: [5]

(a) ____________ solution forms a coloured precipitate with ammonium hydroxide which is soluble in excess of ammonium hydroxide.
[*Ferrous chloride / Copper nitrate*]

(b) Zinc blende is converted to zinc oxide by_________. [*Calcination / Roasting*]

(c) ___________ conducts electricity by the movement of ions.
[*Molten iron / Molten sodium chloride*]

(d) The reaction that takes place at the anode during the electrolysis of aqueous Sodium argentocyanide with silver electrodes is ___________.
[$Ag \rightarrow Ag^+ + e^-$ / $Ag^+ + e^- \rightarrow Ag$]

(e) The salt formed when ZnO reacts with hot concentrated NaOH is ___________. [*sodium zincate / zinc hydroxide*]

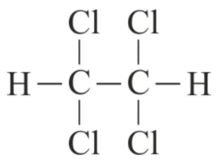
(iv) Match the **Column A** with **Column B**: [5]

	Column A		**Column B**
(a)	$N_2 + 3H_2 \rightleftharpoons 2NH_3$	1.	Vanadium Pentoxide
(b)	$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$	2.	Nickel
(c)	$2SO_2 + O_2 \rightleftharpoons 2SO_3$	3.	Iron
(d)	$C_2H_4 + H_2 \rightarrow C_2H_6$	4.	Concentrated Sulphuric acid
(e)	$CuSO_4.5H_2O \rightarrow CuSO_4 + 5H_2O$	5.	Platinum

(v) (a) Draw the structural diagram for the following organic compounds: **[5]**

1. 2-methyl propene
2. butanal

(b) Give IUPAC name for the following organic compounds:

1.


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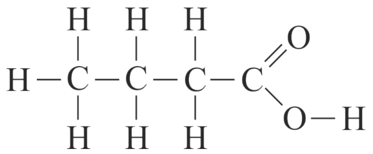
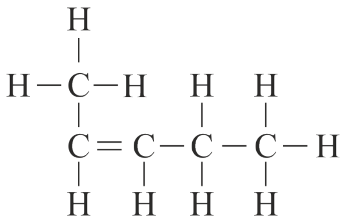


3.



<table>
<tr><th colspan="2">MARKING SCHEME</th></tr>
<tr><td>(i)</td><td>(a) Warm water/ hot water
(b) $AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$
(c) CaO/ Calcium oxide/ quick lime/lime
(d) drying agent/ to absorb moisture/ to remove water / to dry gas
(e) Ammonia gas is soluble in water/ dissolves in water/ highly soluble in water/miscible in water/ forms ammonium hydroxide/ is collected by downward displacement of air / by upward delivery of gas</td></tr>
<tr><td>(ii)</td><td>(a) Spurious / Illicit alcohol
(b) Basic salt/ Basic/ Base
(c) Isomers
(d) Electronegativity
(e) Polar covalent bond/ Polar</td></tr>
<tr><td>(iii)</td><td>(a) Copper nitrate/ Copper (II) nitrate/ $Cu(NO_3)_2$/Cupric nitrate
(b) Roasting
(c) molten sodium chloride/fused NaCl/ Sodium chloride / NaCl/ aqueous NaCl
(d) $Ag \rightarrow Ag^+ + e^-$ / $Ag - e^- \rightarrow Ag^+$
(e) sodium zincate / Na_2ZnO_2</td></tr>
<tr><td>(iv)</td><td>
<table>
<tr><th></th><th>Column A</th><th>Column B</th></tr>
<tr><td>(a)</td><td>$N_2 + 3H_2 \rightleftharpoons 2NH_3$</td><td>3. Iron</td></tr>
<tr><td>(b)</td><td>$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$</td><td>5. Platinum</td></tr>
<tr><td>(c)</td><td>$2SO_2 + O_2 \rightleftharpoons 2SO_3$</td><td>1. Vanadium Pentoxide/ Platinum</td></tr>
<tr><td>(d)</td><td>$C_2H_4 + H_2 \rightarrow C_2H_6$</td><td>2. Nickel / Platinum</td></tr>
<tr><td>(e)</td><td>$CuSO_4 . 5H_2O \rightarrow CuSO_4 + 5H_2O$</td><td>4. Concentrated Sulphuric acid</td></tr>
</table>
(correct number or correct word)
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<tr><th>Column A</th><th>(a)</th><th>(b)</th><th>(c)</th><th>(d)</th><th>(e)</th></tr>
<tr><td>Column B</td><td>3</td><td>5</td><td>1/5</td><td>2/5</td><td>4</td></tr>
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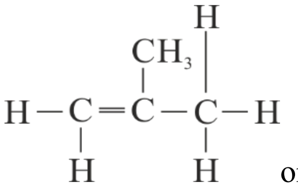
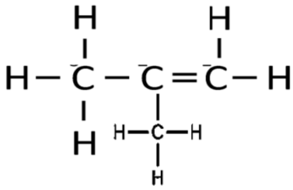
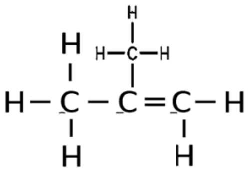
 or  or 

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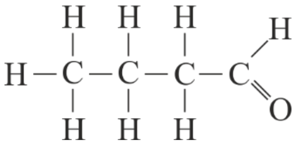


(b) 1. 1, 1, 2, 2 tetra chloroethane

2. butanoic acid/ Butan -1-oic acid

3. 2 - pentene / pent-2-ene

Comments of Examiners

(i) (a) Many candidates correctly identified the substance X as 'warm water'. However, several candidates simply wrote water. A significant number of candidates wrote concentrated sulphuric acid as the answer, indicating confusion about the reagents used.

(b) Many candidates were unable to write the correct balanced chemical equation. Several candidates were confused in writing the correct reactants and products. Several candidates incorrectly wrote 'sulphuric acid' in the reactant part instead of 'water'. Some candidates wrote 'aluminium oxide' as one of the products instead of 'aluminium hydroxide'.

(c) Most candidates correctly identified Y as 'calcium oxide' or 'quick lime'. However, a few candidates mistakenly wrote 'concentrated sulphuric acid' as the 'drying agent'. Some candidates even wrote 'calcium hydroxide' as the incorrect answer.

(d) Many candidates confused 'drying agent' with 'dehydrating agent' and thus answered it incorrectly. Only a few candidates were able to state the correct function of Y.

(e) Many candidates answered it correctly; however, some candidates were unable to explain why 'ammonia' is collected by downward displacement of air and not water. Some candidates were not aware of the high solubility of ammonia in water. A few candidates wrote how the gas is collected without reading the question carefully.

(ii) (a) Many candidates could not recall the correct term 'spurious alcohol' and confused it with 'denatured alcohol' or 'methylated spirit'. Some candidates wrote incomplete or incorrect terms like 'illicit' or 'methylated'.

(b) Only a few candidates with a clear understanding of the concept were able to identify the term 'basic salt' correctly. A significant number of candidates confused it with 'alkaline salt' and 'acidic salt'. Some candidates copied the term 'di acidic base' from the question instead of providing the correct term.

(c) Most candidates answered it correctly, but some confused 'isomer' with 'isomerism'. A few

Suggestions for teachers

- Teach laboratory preparation of ammonia thoroughly, covering reactants, products, drying agents, and conditions of reaction.
- Explain the importance of specific reaction conditions for carrying out the laboratory preparation of ammonia.
- Provide a variety of questions based on the identification of the reagents in the lab setup.
- Emphasise the importance of balanced equations in accurately representing chemical reactions and provide regular practice of writing the correct and balanced equation through quizzes, regular class tests, etc.
- Highlight the need to include all products formed in chemical equations.
- Clearly explain the purpose of the drying agent and make students aware of the specific drying agent used for specific gases, along with the scientific reason.
- Clearly teach the difference between drying agents and dehydrating agents by providing examples of each to avoid confusion while writing the answer.
- Stress on the high solubility of ammonia in water, which is the reason for not collecting it in water.
- Emphasise on collecting ammonia by downward displacement of air and explain the reason for the same.
- Provide worksheets featuring error-spotting exercises—such as identifying the incorrect drying agent or gas collection method to strengthen the conceptual understanding.
- Give more practice on application-level questions, helping students to identify the errors in the diagram along with the reason.
- Ensure that students understand key terms and their precise meanings, emphasising the use of correct words in context.
- Provide ample practice worksheets on salt preparation methods, emphasising examples and key terminologies.
- Explain the meaning of partial replacement of hydroxyl groups and the terms di-basic, di-acidic, and tri-acidic bases, with examples.
- Clarify the difference between acidic and basic salts using demonstrations and

candidates mistakenly wrote 'isotope' or 'homologous series' instead of 'isomer'.

(d) Majority of the candidates wrote the correct term 'electronegativity'. However, some incorrect answers such as 'electron affinity' or 'ionisation potential' were observed. A few used incorrect terms, such as electron negativity or EN, as the answer.

(e) Most candidates correctly identified 'polar covalent bond', but some confused the term with 'non-polar covalent bond' or 'coordinate covalent bonds'.

(iii)(a) A few candidates mistakenly chose 'ferrous chloride' instead of 'copper nitrate', likely due to confusion caused by the formation of precipitates.

(b) Some candidates answered correctly, while others got confused between 'roasting' and 'calcination', leading to incorrect responses.

(c) Many candidates answered, 'molten iron' instead of 'molten sodium chloride' due to a lack of clarity about ionic conduction and electron flow.

(d) Some candidates wrote the incorrect answer, reflecting uncertainty in understanding the correct reaction.

(e) Most candidates correctly identified 'sodium zincate', but some of them incorrectly chose 'zinc hydroxide'.

(iv) Many candidates answered correctly, particularly for the first three parts of the question, identifying 'iron', 'platinum', and 'vanadium pentoxide' as the correct catalysts. However, some candidates got confused in differentiating 'platinum' from 'nickel'. Several candidates also confused concentrated sulphuric acid (H_2SO_4) with other options.

(v) (a1.) Many candidates were able to draw the structure of '2-methyl propene'. However, in several cases, candidates mistakenly drew the 'condensed formula' rather than the 'structural formula' or added an extra hydrogen atom to the carbon with the double bond. Some candidates misplaced the methyl group or incorrectly counted the number of hydrogen atoms, leading to errors in the structure. A few candidates confused the placement of the double bond and the methyl group, resulting in incorrect branching.

(a2.) Many candidates drew the correct structural formula of 'butanal', while others confused –CHO functional group with '–OH'

Suggestions for teachers

- Explain the meaning of the term isomer, highlighting its use with isomerism by giving suitable examples.
- Provide adequate practice of drawing the structural formulas of different organic compounds with the same molecular formula but different structures.
- Emphasise the definitions like electronegativity, electron affinity, etc. that are used in periodic trends.
- Clarify the difference between electronegativity and electron affinity to avoid confusion.
- Teach students the concept of polar and non-polar covalent bonds with correct definitions and appropriate examples.
- Facilitate discussion on roasting and calcination along with the names of the ores that are taken for the same.
- Encourage experimental learning to help students recognise and recall lab observations, especially involving colours and precipitates.
- Demonstrate the chemical reaction of copper nitrate using ammonium hydroxide first in a limited amount and then in excess to differentiate between the precipitates formed.
- Reinforce the difference between ionic and metallic conduction by giving examples of an electrolyte and a metal, respectively.
- Stress the concept of amphoteric oxides, demonstrating their reactions with both acids and alkalis, and encourage students to write balanced equations for these reactions.
- Highlight the OIL RIG concept (Oxidation Is Loss, Reduction Is Gain) and connect it to the reactions occurring at electrodes.
- Ensure that students understand reactions at the cathode and anode based on the loss and gain of electrons.
- Teach the basic concept of catalysts and explain their role in various types of reactions.
- Ask students to make a list of different catalysts and promoters used in both laboratory and industrial processes and make them revise it on a regular basis.

or –'COOH'. Some candidates drew the condensed form of the functional group instead of the structural formula, and some even failed to show the tetravalency of the carbon atom. A few candidates incorrectly placed the –CHO group within the structure or failed to show the double bond of oxygen in 'butanal'.

(b1.) Many candidates correctly wrote the IUPAC name as '1,1,2,2-tetrachloroethane', while some candidates made errors by not mentioning the prefix 'tetra' or incorrectly used terms like 'quadra' or 'quadri' instead of 'tetra'. Additionally, some candidates were unable to give the correct numbering to the chlorine atoms in the structure, and a few of them omitted the numbering.

(b2) Most candidates correctly identified and wrote 'butanoic acid' as the answer. However, some mistakenly wrote 'butanol' instead of the correct answer, while others omitted the word 'acid'. A few candidates either made spelling errors or did not correctly number the carbon atoms in the structure.

(b3) Many candidates confused the IUPAC naming and mistakenly wrote 'pentane' instead of the correct answer 'pent-2-ene'. Some candidates were able to identify the organic compound correctly. A common error observed was incorrectly identifying the longest carbon chain, which led candidates to write '1-methylbut-1-ene' and lose marks.

Suggestions for teachers

- Train students to draw structural formulae properly, ensuring that all carbon atoms satisfy tetravalency.
- Elucidate clearly the IUPAC rules for naming and drawing organic compounds particularly for compounds with functional groups.
- Conduct peer group activity where one group can suggest an unfamiliar compound and the other group names it following the IUPAC rules.
- Explain the difference between structural and condensed formula and ensure thorough practice of the same to improve conceptual clarity.
- Facilitate revision of functional groups with the help of flashcards for quick recall.
- Emphasise the importance of correct placement of substituents or functional groups according to IUPAC rules.
- Provide ample practice with worksheets for naming and drawing organic compounds.
- Teach students to identify the longest carbon chain in organic compounds and number the carbons correctly.
- Elucidate that the functional group should be given the lowest numbering in the carbon chain.
- Clarify how branching and the length of the carbon chain affect naming of the organic compound.
- Encourage regular practice of drawing structures for various organic compounds.

SECTION B (40 Marks)

*(Attempt **any four** questions from this **Section**.)*

Question 3

(i) The atomic number of two atoms **'X'** and **'Y'** are 14 and 8 respectively. **[2]**

State:

(a) the period to which **'X'** belongs.

(b) the formula of the compound formed between **'X'** and **'Y'**.

*(Do not identify **X** and **Y**)*

(ii) Justify the following statements: **[2]**

(a) Anode is known as the oxidizing electrode.

(b) Graphite electrodes are preferred in the electrolysis of molten lead bromide.

(iii) The reaction between concentrated sulphuric acid and magnesium can be represented by the equation given below: **[3]**

$$Mg + 2H_2SO_4 \rightarrow MgSO_4 + 2H_2O + SO_2$$

If 60 g of magnesium is used in the reaction, calculate the following:

(a) The mass of sulphuric acid needed for the reaction.

(b) The volume of sulphur dioxide gas liberated at S.T.P.

[*Atomic weight: Mg=24, H=1, S=32, O=16*]

(iv) Give one **significant** observation when: **[3]**

(a) a solution of barium chloride is added to zinc sulphate solution.

(b) lead nitrate is heated in a test tube.

(c) chlorine gas is passed over moist starch iodide paper.

MARKING SCHEME

(i)	(a) Period 3/ third period/ 3/third (b) XY_2/ Y_2X/X_2Y_4 / Y_4X_2
(ii)	(a) Anions or ions migrate to the anode and give up / lose electrons / donate electrons (and get oxidised) / Anode receives/ gains electrons (causing oxidation)/ Oxidation takes place (b) Graphite electrodes are not attacked/ remains unaffected by (Br_2 vapours)/ Graphite is an inert /unreactive/non-reactive electrode
(iii)	(a) G.A.W of Mg =24 G.M.W of H_2SO_4= 2+ 32+ (4x16) =98g/98 24g of Mg needs 2 x 98g of sulphuric acid $\therefore$60g of magnesium will need x g of sulphuric acid X = 2 x 98 x 60/24 = 490 g of sulphuric acid/ 490 (b) 24g of magnesium will liberate 22.4L of Sulphur dioxide $\therefore$60 g of magnesium will liberate y L of Sulphur dioxide. Y = 22.4 x 60/24 = 56 L of Sulphur dioxide/ 56/56000cc
(iv)	(a) White ppt / white insoluble substance/ white substance is formed (b) Reddish brown gas / buff/ yellow residue / a gas which rekindles a glowing splinter/ crackling sound is heard/ decrepitation takes place/ a solid which fuses to the glass (c) Starch iodide paper turns blue black/ bluish black

Comments of Examiners

(i) (a) Most candidates correctly identified the period. A few wrote the period number in Roman numeral (III) instead of Arabic numeral (3), leading to loss of marks.

(b) Most candidates correctly wrote the formula as 'XY_2' demonstrating an understanding of the elemental ratio. However, some candidates mistakenly provided the name of the compound instead of the formula. A few candidates mentioned the formula such as 'X_2Y' or 'X_6Y_{12}' which was incorrect.

(ii) (a) A significant number of candidates failed to answer this question correctly. Many candidates wrongly stated that the anode loses electrons instead of mentioning that anions migrate towards anode and lose electrons. Some candidates got confused between the concepts of 'oxidation' and 'reduction' and related it to the gain or loss of oxygen/hydrogen instead of 'electron transfer'.

(b) Many candidates wrote the correct answer, but some candidates failed to provide the appropriate reasoning for the preference of graphite.

(iii) (a) Most of the candidates correctly calculated the mass of sulphuric acid though some of them wrote the final answer as 98 g without showing calculations. Some candidates lost marks either due to calculation error or substituting the incorrect values in molar mass.

(b) Most of the candidates correctly calculated the volume of gas. However, some candidates were confused in calculating the volume at S.T.P. and substituted the values incorrectly while calculating the volume. A few candidates only mentioned the final answer without showing the steps for calculation.

(iv)(a) Most candidates answered the question correctly by identifying the formation of a 'white precipitate' or a 'white insoluble substance'. However, some candidates wrote either the chemical equations or products formed rather than clearly stating the observations. A few of them were confused between the colour 'white' and 'blue'.

Suggestions for teachers

- Explain the definitions of period and group in the periodic table.
- Facilitate thorough revision of the periodic table emphasising on the characteristics of different groups and periods.
- Teach students to find the group number and period number with the help of atomic number via. electronic configuration.
- Ensure students write the period number in Arabic numerals only.
- Integrate the concepts of atomic structure and chemical bonding in periodic table to write the formula of a compound.
- Provide repetitive practice of writing accurate chemical formulas based on criss-cross method and reinforce valency matching with plenty of examples.
- Clarify oxidation-reduction concepts and the roles of oxidising and reducing agents using equations.
- Relate the concept of oxidation and reduction putting stress on the fact that oxidation occurs at the anode and reduction occurs at the cathode.
- Use visual stimuli like animations or short videos to show the movement of electrons and changes occurring at each electrode during electrolysis.
- Highlight that inert electrode, such as graphite or platinum, do not participate in the electrolytic reaction and remain unchanged.
- Guide students to read numerical questions carefully, understand what is being asked and then solve the problem in a stepwise manner.
- Provide ample practice of questions related to calculation of molar mass of a given compound.
- Clearly highlight that the equation must be balanced before calculating the volume or the mass of the given compound.
- Reinforce the distinction between weight and volume in the context of Gay-Lussac's Law and offer adequate practice of the same.

(b) Most candidates scored marks by correctly stating the observations. However, a few could not answer accurately. Some candidates only mentioned 'brown gas' instead of specifying the colour of the gas. Some wrote 'NO_2 is formed' without describing its colour. Few mentioned it to be a 'colourless gas'.

(c) Most candidates answered it correctly although incorrect observations were found in some answer scripts.

Suggestions for teachers

- Focus on practicing mole concept problems and teach how to substitute values in the correct order.
- Explain the use of standard molar volume (22.4 L/mol or 22400 c.c/mol) in calculations, emphasising direct formula application.
- Emphasise learning chemical reactions through practical experiments wherever possible to help students recognise and recall observations effectively.
- Provide thorough revision of observation-based questions, highlighting the importance of detailed observations (e.g., colour changes, precipitate formation, gas evolution).
- Demonstrate double displacement reactions, both with and without precipitation, using solubility charts.
- Encourage students to justify their observations with reasoning and confirmatory tests.
- Highlight the difference between observation and inference clearly.
- Conduct experiments and use visual aids or videos to demonstrate actual colour changes, gas evolution, and other observations in the given reactions.
- Ensure students learn and remember the correct colour of gases by asking oral questions in the class.
- Guide students to write exact initial and final colour of the reactant and product formed, if there is a colour change.

Question 4

(i) A gas cylinder can hold 150g of hydrogen under certain conditions of temperature and pressure. If an identical cylinder with the same capacity can hold 450g of gas **'G'** under the same conditions of temperature and pressure, find: **[2]**

(a) the vapour density of the gas **'G'**.

(b) the molecular weight of gas **'G'**.

(ii) Complete and balance the following equations: **[2]**

(a) $CH_3COONa + NaOH \xrightarrow[\Delta]{CaO}$

(b) $CH_3COOH + Mg \rightarrow$

(iii) Name the **gas** produced during each of the following reactions: **[3]**

(a) When copper is treated with hot, concentrated nitric acid.

(b) When ammonia is burnt in an atmosphere of oxygen.

(c) When ferrous sulphide reacts with dilute hydrochloric acid.

(iv) Study the table given below. Use only the letters given in the table to answer the questions. **Do not identify** the elements. **[3]**

IA	IIA	IIIA	IVA	VA	VIA	VIIA	0
			E		J		Q
	L			G			
	M	D				P	
	N						

(a) State the valency of element 'G'.

(b) Which element can exhibit catenation?

(c) Write the formula of the compound formed between 'M' and 'P'.

MARKING SCHEME

(i)	(a)	Vapour density = $\frac{450}{150}$ or only 3
	(b)	Molecular weight = 3 x 2 = 6
(ii)	(a)	$CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$
	(b)	$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$
(iii)	(a)	Nitrogen dioxide /NO_2/
	(b)	Nitrogen /N_2
	(c)	Hydrogen sulphide / H_2S
(iv)	(a)	3/ three
	(b)	E
	(c)	MP_2

Comments of Examiners

(i) (a) Many candidates correctly calculated the vapour density as 450/150 = 3. Some lost marks either due to confusion regarding the relationship between vapour density and molecular weight or were unaware of the formula. A few candidates made calculation errors, while others provided the direct answer without the calculations.

(b) Most of the candidates calculated the molecular weight correctly. However, a few candidates simply mentioned the final answer as '6' without showing the calculations.

(ii) (a) Most of the candidates correctly wrote the products. However, some candidates wrote '$NaHCO_3$' as one of the products instead of 'Na_2CO_3', while a few others mentioned the incorrect products or provided the unbalanced equation.

(b) Majority of the candidates could not balance the given chemical equation. Only a few candidates were able to write the correct balanced equation.

(iii)(a) Many candidates correctly answered it as 'nitrogen dioxide'. However, a significant number of candidates confused 'nitrogen dioxide' with 'nitric oxide'. Some candidates even wrote 'hydrogen gas'.

(b) Several candidates correctly identified the gas as 'nitrogen'. However, a few candidates confused it with 'nitrogen dioxide' or 'nitric oxide'.

(c) Most candidates correctly named the gas as 'hydrogen sulphide', but a few candidates wrote 'sulphur dioxide' instead.

(iv) (a) Many candidates stated the correct valency of the element as '-3', while some candidates mentioned only the numeric value without including the sign. A few candidates even wrote the valency as '5', due to confusion between valency and the number of valence electrons.

(b) Majority of the candidates were able to score marks in this part of the question. However, a few mentioned the element as carbon instead of the designated letter 'E'.

(c) Most candidates wrote the correct formula as 'MP_2'. However, a few of them mentioned M to be hydrogen in the formula and wrote 'HP_2'. Some candidates reversed the order of the elements and wrote 'P_2M' for which no marks were awarded.

Suggestions for teachers

- Emphasise using the correct formulas and performing accurate calculations, followed by rechecking of the answers while solving numerical problems.
- Provide rigorous practice and discussion of completing and balancing equations of alkanes, alkenes, and alkynes.
- Instruct students to mention the name and not the formula of the gas if the question demands the name of the gas produced.
- Emphasise the product formed when copper reacts with concentrated HNO_3 and dilute HNO_3 using practical demonstrations for better understanding and retention.
- Highlight the difference between the reaction of concentrated and dilute nitric acid in terms of the evolution of NO_2 and NO, respectively, along with the reason. Prompt the students to think more about 'why' and 'what if' dimensions in each of these scenarios.
- Ensure that students learn the name and the colour of the gas produced during a reaction wherever possible.
- Facilitate practice and revision of the chemical properties of ammonia, with emphasis on writing and balancing the equations.
- Explain the reaction of dil. hydrochloric acid with metallic sulphides and demonstrate it in the laboratory to identify the rotten egg-smelling gas.
- Provide ample practice on periodic trends across periods and down the groups, focusing on properties based on atomic number and nuclear charge.
- Clearly explain the relationship between group number and number of outermost electrons, and how to determine valency from electronic configuration.
- Instruct students not to identify elements by name but to use the provided alphabetical symbols in the question.
- Emphasise the periodic trend of properties of elements as they move across a period from left to right, relating to their electronic configuration.

Suggestions for teachers

- Explain the meaning of the term catenation and the group in the periodic table that can exhibit catenation.
- Familiarise students with the type of questions where students must identify the element along with writing the formula, through regular practice.
- Instruct students to write the metallic ion first, followed by the non-metallic ion while writing the chemical formula.

Question 5

(i) Given below are two sets of elements from two different periods. **[2]**

Name the element with the **highest** ionisation potential in each of the following sets.

(a) Al, Cl, Mg

(b) Ne, O, F

(ii) Ammonia gas is passed over heated copper (II) oxide in a combustion tube: **[2]**

(a) Name the gas evolved.

(b) What will be the colour of the residue that is left in the combustion tube at the end of the reaction?

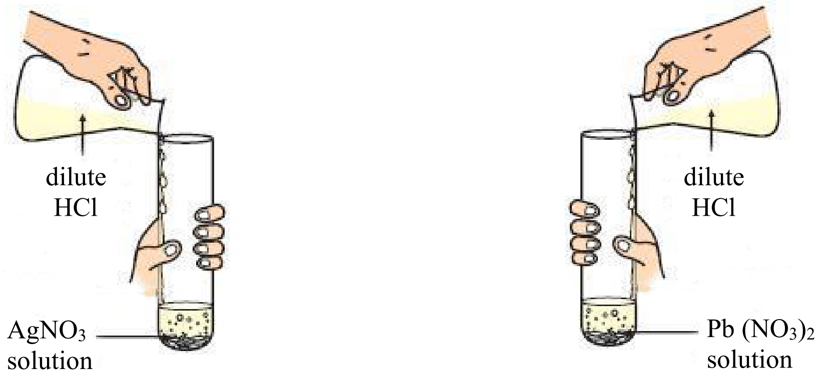
(iii) Give balanced equations for the following: **[3]**

(a) Action of dilute hydrochloric acid on ammonium carbonate.

(b) Oxidation of sulphur with hot concentrated nitric acid.

(c) Reaction of concentrated sulphuric acid with carbon.

(iv) Rohit took two different salt solutions in test tubes **C** and **D** as shown in the figure below. He added dilute HCl to each of the two test tubes. The products formed in the test tubes **C** and **D** are *silver chloride* and *lead chloride* respectively. **[3]**



Test tube **C** Test tube **D**

State:

(a) one common observation made by Rohit in both the reactions.

(b) the observations made by him on addition of excess of ammonium hydroxide to the products formed in:

1. test tube **C**
2. test tube **D**

MARKING SCHEME

(i)	(a) Cl/ chlorine/Cl_2 (b) Ne/ Neon
(ii)	(a) Nitrogen / N_2 (b) Pink / reddish brown/ brown/ pinkish brown/ brownish pink
(iii)	(a) $(NH_4)_2CO_3 + 2HCl \rightarrow 2NH_4Cl + H_2O + CO_2$ (b) $S + 6HNO_3 \rightarrow H_2SO_4 + 2H_2O + 6NO_2$ (c) $C + 2H_2SO_4 \rightarrow CO_2 + 2H_2O + 2SO_2$
(iv)	(a) white ppt / ppt / insoluble substance / white substance formed (b) In C (white) ppt dissolves/ is soluble in excess/ forms clear or colourless solution/ forms solution In D (white) ppt does not dissolve/ remains insoluble/ remains unchanged/ (white) ppt remains/ no change/ white ppt

Comments of Examiners

(i) (a) Most candidates correctly identified the element with the highest 'ionisation potential' as 'chlorine' (Cl). However, some candidates either arranged the elements without mentioning the name of the element or used the wrong notation. A few confused 'ionisation potential' with 'electronegativity'.

(b) Majority of the candidates correctly identified the element as 'neon' (Ne). However, a few candidates incorrectly wrote 'fluorine' (F).

(ii)(a) 'Nitrogen gas' was correctly identified by many candidates. However, a few were confused and incorrectly identified the gas as nitrogen dioxide (NO_2), nitric oxide (NO), hydrogen (H_2), or even water (H_2O), indicating a lack of clarity about the reaction.

(b) The majority of candidates correctly identified the colour of the residue as 'reddish-brown'. However, some candidates were confused with the colour of the residue and wrote random colours like blue, pink, brown or simply black.

(iii)(a) Most of the candidates wrote the unbalanced. equation or the incorrect products. However, few candidates were able to write the correct products. A significant number of candidates mentioned 'sulphuric acid' as one of the products instead of 'carbon dioxide' and 'water'.

(b) Many candidates could not write the correct products for the given reactants. Although some candidates wrote the correct chemical equation, balancing it accurately posed a challenge to them.

(c) Most candidates answered this correctly by writing the balanced chemical equation. However, some candidates failed to balance the equation.

(iv)(a) Majority of the candidates correctly stated the observation that 'white precipitate' is formed. Some candidates confused the observation with the product formation.

(b) (1) Many candidates correctly identified the solubility of the precipitate by stating that it dissolves in excess ammonium hydroxide. Some candidates incorrectly stated that the precipitate is 'insoluble' instead of 'soluble'.

(b) (2) Many candidates correctly stated that the white precipitate does not dissolve in excess ammonium hydroxide. Some candidates gave

Suggestions for teachers

- Thoroughly explain the variation in periodic properties across periods and down groups, covering all periodic trends like ionisation potential (IP), electron affinity (EA), and electronegativity (EN).
- Emphasise the correct usage of '<' and '> 'signs to represent the trends accurately.
- Instruct students to read questions carefully and pay attention to the word highlighted in bold.
- Reinforce the concept of periodic trends by providing practice worksheet.
- Use step-by-step reaction mechanism to illustrate key concept such as:
  - Ammonia acting as a reducing agent.
  - Reduction of CuO to Cu.
  - Formation of nitrogen as a by-product.
- Provide clear observations for each reaction by real laboratory demonstrations or through virtual lab platforms available online.
- Train students to read the question carefully to understand exactly what is being asked—especially when marks are awarded specifically for observations.
- Guide students to always balance a chemical reaction and mention the name of the reactants and products formed.
- Use reaction-type templates to help students recognise patterns, such as:
  - Acid + Carbonate → Salt + CO_2 + H_2O
  - Acid + Base → Salt + H_2O
  - Metal + Acid→ Salt + H_2
- Review student responses carefully during practice sessions to identify common misconceptions or errors.
- Use group-wise solubility rule charts and summary tables to help students classify and remember precipitates and their characteristics.
- Emphasise both theory and practical components of analytical chemistry by integrating lab experiments with classroom explanations.
- Encourage students to write detailed observations, including colour, state, and confirmatory tests, rather than just naming the products.

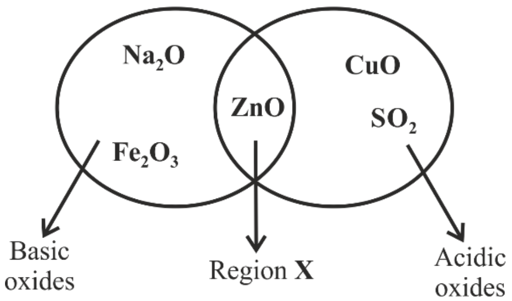
incomplete observations by only mentioning colour change without specifying the solubility of the precipitate. Some other candidates either misunderstood the solubility rule or wrote a chemical equation instead of an observation.

Suggestions for teachers

- Teach the concept of using reagents in both limited and excess quantities to help students understand the formation and dissolution of precipitates.
- Reinforce the concept of precipitation reaction through worksheets and oral quizzes with emphasis on correct sequence (add in small amount → observation → add in excess → observation).
- Instruct the students to write the accurate and complete observation statements, such as: 'white precipitate forms, then dissolves in excess.'
- Conduct hands-on experiments to show the precipitation reaction and changes in solubility and colour of the precipitate when reagent is added in excess.

Question 6

(i) Given below is a diagram showing the placement of five different oxides. With respect to the given diagram answer the following questions: **[3]**



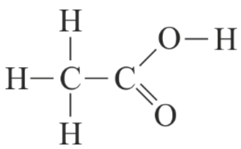
(a) Name the **type** of oxide represented in region **X** in the diagram.

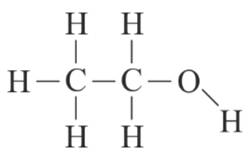
(b) Identify the oxide which has been **incorrectly** placed in the above diagram.

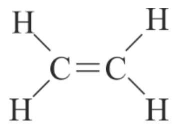
(c) Name the oxide from the above diagram which will form an **alkali** when dissolved in water.

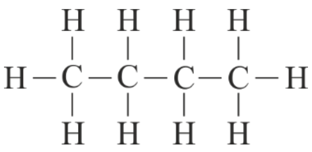
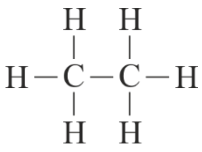
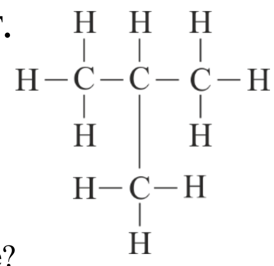
(ii) Given below are organic compounds labelled **A** to **F**. **[3]**

Answer the questions that follow:

A.


B.


C.


D.  **E.**  **F.** 

(a) Which compound forms a **single** product with **bromine**?

(b) Which **two** compounds have the **same** molecular formula?

(c) Which **two** compounds will react together in the presence of concentrated H_2SO_4 to form a product with a **fruity smell**?

(iii) An organic compound '**X**' contains carbon, oxygen and hydrogen only. The percentage of carbon and hydrogen are 47.4% and 10.5% respectively. The relative molecular mass of **'X'** is 76. Find the **empirical** formula and the **molecular** formula of **'X'**. **[4]**

[*Atomic weight: C = 12, O = 16, H = 1*]

MARKING SCHEME

(i)
(a) Amphoteric oxide
(b) CuO/ Copper oxide/ Copper(II) oxide/ Cupric oxide/ Cu
(c) Na_2O / Sodium oxide/Na

(ii)
(a) C / C_2H_4 / ethene/ Ethylene or correct structure
(b) D and F/ butane & 2- methyl propane/ butane & iso- butane or correct structures
(c) A and B/ Ethanoic acid & Ethanol/Acetic acid & Ethyl alcohol or correct structures

(iii)

	C	O	H
% by mass	47.4	42.1	10.5
Ar	12	16	1
No. of moles	47.4/12 = or 3.95	42.1/16 = 2.63	10.5/1 = 10.5
÷ by smallest no.	3.95/2.63 = 1.5	2.63/2.63 = 1	10.5/2.63 = 4
Ratio	1.5 x 2 = 3	1 x 2 =2	4 x 2 = 8

Empirical Formula = $C_3O_2H_8$

n=1

Molecular Formula = $C_3O_2H_8$

Comments of Examiners

(i) (a) Most of the candidates gave the correct answer as 'zinc oxide'. However, some candidates confused 'amphoteric oxide' with 'basic oxide' or 'neutral oxide'.

(b) Most of the candidates could identify the correct answer, while some confused the correct answer, which is 'CuO', with 'Na_2O', and a few of them incorrectly wrote 'Fe_2O_3'.

(c) Many candidates correctly identified that 'Na_2O' dissolves in water to form an alkali, but a significant number confused it with 'CuO'. However, some candidates could not recall the 'solubility rule'.

(ii) (a) The majority of the candidates identified the correct answer as 'C' while a few of them mistakenly chose 'E' either due to confusion in identifying the 'functional groups' involved or misunderstanding the 'structural isomers' as 'chain isomers'. Some candidates did not attempt this portion at all.

(b) Many candidates were able to choose the correct option for the pair of compounds. However, a few of them failed to identify the correct pair and ended up selecting the incorrect options.

(c) Most candidates correctly identified the pair of compounds, although some confused it with options A and C instead of A and B, which led to a loss of marks.

(iii) Many candidates were able to determine the empirical formula but failed to calculate the molecular formula. Some candidates mistakenly divided atomic weights by percentage values and got the incorrect number of moles. Some of the common mistakes observed were:

- Not dividing the number of moles obtained with the smallest number of mole.
- Not multiplying the 'empirical formula' with 'n' to get the 'molecular formula'.

Suggestions for teachers

- Explain the concepts of acidic, basic, amphoteric, and neutral oxides with equations, and highlight the differences between them with suitable examples of each category.
- Thoroughly explain the solubility rules for metal oxide.
- Give practice exercise in class to identify the salts that are soluble or insoluble in water.
- Clearly explain the properties of alkenes and encourage the students to write the correct balanced equations for addition reaction.
- Teach the concept of isomerism and isomer clearly by providing sufficient examples.
- Explain esterification reaction with special reference to the characteristics of the product formed.
- Clarify that the empirical formula represents the simplest whole-number ratio of atoms in a compound.
- Lay stress on the formula that relates empirical formula weight and molecular weight of a compound.
  Molecular weight = n x Empirical formula weight
- Provide classroom practice of numerical problems based on empirical formula, instruct them to show the working and discuss in detail the common errors made by the students.

Question 7

(i) Seema added a few pieces of copper turnings to a test tube containing concentrated acid **P** and she noticed that a reddish-brown gas evolved. **[2]**

(a) Name the acid **P** used by Seema.

(b) Write a balanced chemical equation for the reaction that took place.

(ii) Answer the following questions with reference to the concentration of **bauxite ore**. **[2]**

(a) Name the process used to concentrate the ore.

(b) Give a balanced chemical equation for the conversion of aluminium hydroxide to pure alumina.

(iii) Draw the **dot and cross** structure of the following: **[3]**

(a) An ionic compound formed when Mg reacts with the dilute HCl.

(b) A covalent compound formed when H_2 reacts with Cl_2.

(c) The positive ion produced when ammonia gas is dissolved in water.

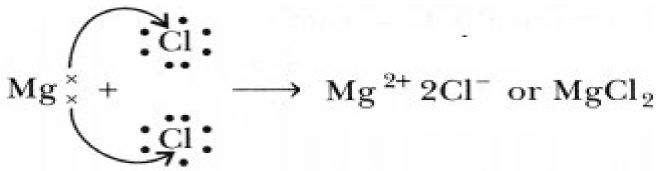
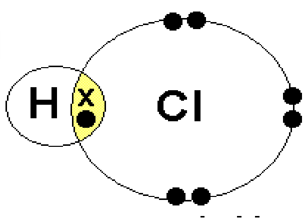
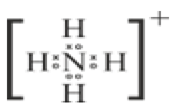
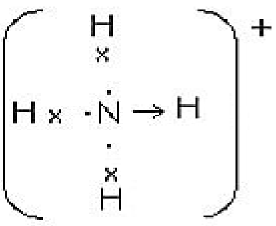
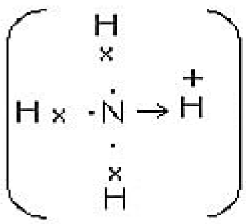
[*Atomic number: Mg = 12, Cl = 17, H = 1, N = 7*]

(iv) Acidulated water is electrolysed using platinum electrodes. **[3]**
Answer the following questions:

(a) Why is dilute sulphuric acid added to water?

(b) Write the reaction taking place at the cathode.

(c) What is the observation at the anode?

MARKING SCHEME	
(i)	(a) concentrated HNO_3 / nitric acid/ aqua fortis (b) $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$
(ii)	(a) Baeyer's Process/ Hall's process/Leaching (b) $2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$
(iii)	(a)  Or  or  (b)  Or  (c)  or  or 

(iv)	(a) (Being covalent) water does not ionize/water is a bad or poor or non-conductor of electricity/ to increase conductivity / acts as a catalyst /speeds the rate of reaction/ water will ionize/ to produce (free mobile) ions/ water is a non-electrolyte (b) $H^+ + 1e^- \rightarrow [H]$ / $2H^+ + 2e^- \rightarrow H_2$ (c) bubbles of colourless gas released/ Volume of gas collected is almost half of that collected at cathode/ gas released rekindles glowing splinter/ gas released turns alkaline pyrogallol solution brown

Comments of Examiners

(i) (a) Most candidates correctly identified the concentrated acid as 'HNO_3', although a few wrote 'HCl' or 'H_2SO_4' and lost marks. Some candidates simply wrote nitric acid, without mentioning the word 'concentrated'.

(b) Majority of the candidates wrote the correct balanced equation, although some failed to balance it. A few candidates mistakenly wrote 'copper oxide' instead of 'copper nitrate' as a product. Others wrote 'NO' instead of 'NO_2' on the product side.

(ii) (a) Most candidates wrote the correct name of the process as 'Baeyer's process'. However, a few candidates failed to answer correctly, confusing the 'Baeyer's process' with 'Hall-Heroult's process'.

(b) Many candidates successfully balanced the chemical equation for the conversion of 'aluminum hydroxide' to 'alumina'. However, some confused the steps of the Baeyer's process and included multiple reactions. Only a few candidates were unable to balance the chemical equation.

(iii)(a) Many candidates drew the correct dot and cross structure for $MgCl_2$, accurately showing the two electrons transferred from Mg to Cl.

However, some common errors observed were:

- Missing out charge on the ions.
- Depicting covalent bonding instead of ionic bonding between Mg and Cl.
- Assigning the incorrect charge on chlorine.
- Failure to differentiate between Mg from Cl as both were represented by a dot.
- Drawing orbital structure instead of the dot and cross structure.

(b) Many candidates correctly drew the dot-and-cross structure of HCl, showing the shared pair of electrons between hydrogen and chlorine. A common error was omitting the lone pairs on chlorine, leading to the incomplete structure.

(c) Many candidates were unable to draw the correct dot-and-cross structure of the NH_4^+ ion, though some did it correctly. Some common errors observed were:

- Missing positive charge on ammonium ion.

Suggestions for teachers

- Emphasise the reaction of nitric acid with copper, producing NO_2 or NO depending on the concentration of the acid and the reaction conditions.
- Facilitate observation-based learning while teaching the properties of both concentrated and dilute nitric acid.
- Clarify that conc. nitric acid, when reacts with metal, releases reddish brown gas while dil. nitric acid releases nitric oxide which is colourless.
- Highlight on how reaction conditions affect the product formed, especially the nature of the gas evolved.
- Explain explicitly the steps involved in the extraction of metal from its ore.
- Teach the correct names and functions of each process involved in the concentration of bauxite ore.
- Stress upon the representation rules for electron dot structures.
- Reiterate to use distinct symbols while drawing electron dot structure of the compounds involving two different elements: e.g., dots (•) for one element and crosses (×) for another element.
- Provide rigorous practice of drawing the electron dot diagrams neatly through worksheets, blackboard tests, etc.
- Highlight the common mistakes that students make in drawing these types of electron dot structures.
- Explicitly teach the concept of coordinate bonding by using examples like bonding in ammonium ion (NH_4^+) and hydronium ion (H_3O^+).
- Ensure that students must represent the coordinate bond clearly, i.e. an arrow from the donar pointing towards the acceptor in the coordinate compounds.
- Emphasise that ion availability is key to electrical conductivity in solution.
- Demonstrate electrolysis of acidulated water in the lab with special mention of the role of sulphuric acid, platinum electrodes, and gases evolved at respective electrodes.
- Encourage students to observe the evolution of gases at the electrodes during the demonstration.

- Non-inclusion of the lone pair on nitrogen.
- Showing a bonded structure instead of the dot-and-cross representation,
- Depicting lone pairs and arrows simultaneously.

(iv) (a) Many candidates wrote the correct reason for adding 'dilute H_2SO_4' to water. Some of them could not explain it properly and only mentioned water being a polar covalent compound without elaborating why H_2SO_4 is necessary. Some candidates gave vague answers.

(b) Most of the candidates wrote the correct half-equation; however, many of them wrote as $4H^+ + 4e^- \rightarrow 2H_2$ instead of $2H^+ + 2e^- \rightarrow H_2$. Some candidates were unable to provide a reaction at the cathode.

(c) 'Oxygen gas' released at the anode was correctly identified by most of the candidates, but some candidates only wrote 'O_2 is evolved' without mentioning the confirmatory test for it.

Suggestions for teachers

- Familiarise students with the confirmatory tests for the gases that evolved during electrolysis of acidulated water.
- Provide regular practice of writing the observations during the electrolysis through class tests or revision assignments.

Question 8

(i) (a) State Avogadro's Law. **[2]**

(b) Define Co-ordinate bond.

(ii) Differentiate between the following pairs of compounds using the **reagent** given in the bracket: **[2]**

(a) Ammonium chloride and Sodium chloride *(using an alkali)*

(b) Zinc Nitrate solution and Calcium Nitrate solution *(using excess sodium hydroxide solution)*

(iii) You are provided with some compounds in the box. **[3]**

PbO	CH_4	PbO_2	CO_2
	HCl	NCl_3	SO_2

Choose the most appropriate compound which fits the descriptions (a) to (c) given below:

(a) A colourless gas which turns acidified $K_2Cr_2O_7$ from *orange to green*.

(b) A *yellow explosive* oily liquid formed when excess chlorine gas reacts with ammonia gas.

(c) A *yellow metallic* oxide formed on thermal decomposition of $PbCO_3$.

(iv) **P**, **Q**, **R** and **S** are the different methods of preparation of salts. **[3]**

P – Simple displacement

Q – Neutralisation by titration

R – Precipitation

S – Direct combination

Choose the **most appropriate** method to prepare the following salts:

(a) $PbCl_2$

(b) $FeCl_3$

(c) Na_2SO_4

MARKING SCHEME

(i)	(a)	Avogadro's Law: Under the **same** conditions of temperature and pressure **equal** volume of all gases contain the same number of molecules/ particles.
	(b)	Co-ordinate bond: The bond formed between two atoms by sharing a pair of electrons provided entirely by one of the combining atoms.
(ii)	(a)	NH_4Cl reacts with alkali to produce pungent colourless gas / or a gas which turns **moist** red litmus paper blue/ any test for NH_3 …. Dense white fumes with conc. HCl / turns Nessler's reagent brown/ turns solutions of copper salt into blue ppt or deep inky blue solution(excess)/ gas evolved With NaCl - no reaction/ no gas evolved/ no visible change or reaction/ no such observation
	(b)	Zinc nitrate gives a (gelatinous)(white) precipitate which is soluble in excess of NaOH/ppt dissolves in excess of NaOH/ clear solution formed Calcium nitrate gives a (white) ppt which is insoluble in excess of NaOH /ppt does not dissolves in excess of NaOH/ white ppt remains / sparingly soluble
(iii)	(a)	SO_2 / Sulphur dioxide
	(b)	NCl_3 / Nitrogen trichloride
	(c)	PbO / Lead oxide/ lead monoxide/ Litharge
(iv)	(a)	Precipitation / R
	(b)	Direct combination / S
	(c)	Neutralisation by titration / Q

Comments of Examiners

(i) (a) Most candidates stated Avogadro's law correctly; however, a few of them missed out the key phrase – 'under the same temperature and pressure'.

(b) Many candidates were unable to define 'coordinate bond' correctly and confused 'lone pairs' with 'shared pairs' of electrons.

(ii) (a) Many candidates failed to differentiate between 'ammonium chloride' and 'sodium chloride' using an alkali, as they either wrote the reactions for both or missed the key confirmatory tests for 'ammonia'. Several candidates just mentioned 'a colourless gas is formed' without mentioning 'ammonia' or the confirmatory test. A common error observed was specifying that 'ammonium chloride' reacts with an 'alkali' to give 'ammonia gas', which turns moist red litmus blue, while 'sodium chloride' does not react.

(b) Many candidates were unable to differentiate between 'zinc nitrate' and 'calcium nitrate' using excess 'sodium hydroxide' solution. Several candidates were confused and wrote either 'both are soluble' or 'both are insoluble'. Some answered it correctly that 'zinc nitrate' forms a white precipitate soluble in excess of 'sodium hydroxide', whereas 'calcium nitrate' forms a white precipitate that is insoluble in excess of 'sodium hydroxide'. A few candidates wrote the reaction for only one of the salts.

(iii)(a) Many candidates chose the correct answer as 'SO_2', while a few incorrectly chose 'CO_2' or 'CH_4'.

(b) Many candidates correctly identified the yellow explosive oily liquid as 'NCl_3', while others failed to do so as they missed out the word 'excess'.

(c) Most candidates correctly identified the yellow metallic oxide as 'PbO', while some of the candidates wrote the incorrect answers such as 'PbO_2' or 'ZnO'.

(iv)(a) Many candidates incorrectly chose method 'P' instead of method 'R'. Some candidates randomly chose the method out of the given methods.

(b) Majority of candidates answered it correctly, but some candidates lost marks for selecting the incorrect method for the preparation of 'ferric chloride'.

(c) Most candidates identified the most appropriate method to be 'neutralisation by titration', though a few candidates confused it with 'simple displacement'.

Suggestions for teachers

- Emphasise the correct definitions and Laws with keywords and conditions.
- Demonstrate differentiating the pairs of compounds using the reagent in the lab.
- Lay stress on observing the precipitate if formed while adding the reagent to the compound and its solubility in limited and excess use of the reagent.
- Acquaint students with examples where they must differentiate between the two compounds using the same reagent.
- Teach differentiation between colourless gases like ammonia, carbon dioxide and hydrogen through confirmatory tests.
- Emphasise the key chemical test to distinguish sulphur dioxide from carbon dioxide, which is the acidified potassium dichromate test.
- Highlight that carbon dioxide does not affect the acidified potassium dichromate solution.
- Use animations or simulations to show how the product changes based on the relative amounts of NH_3 and Cl_2.
- Display a solubility chart in the classroom for constant reference.
- Use animations or videos to demonstrate mixing of salt solutions and formation of precipitate.
- Conduct lab demonstrations wherever possible for visual memory.
- Provide worksheets based on writing equations, observations, and identifying the precipitate.
- Give MCQs based on salt names and their preparation methods.

GENERAL COMMENTS

Topics found difficult by candidates	
	• Assertion and Reason questions. • Writing the correct equations, formulas and balancing the equations. • Understanding and applying the mole concept (including empirical and molecular formulae). • Drawing dot and cross structures in chemical bonding. • Practical chemistry questions related to observation and confirmatory tests. • Identifying the correct method for the preparation of salts. • Numerical based on Gay-Lussac's Law, weight-volume relationships, stoichiometry. • IUPAC nomenclature and structural formulae of the organic compounds. • Understanding cathode/anode reactions in electrolysis. • Answering application-based and observation-based questions.

Concepts in which candidates got confused	
	• Determination of empirical and molecular formula. • Difference between dot-and-cross structures and the orbital structures of the compound. • Proper IUPAC nomenclature and drawing structural formulae for organic compounds. • Differentiating organic structures and identifying the longest carbon chain. • Distinguish between dehydrating agents and drying agents. • Balancing chemical equations. • Representing lone pairs and shared pairs of electrons correctly in the diagram. • Identification of substances through confirmatory tests and observations. • Understanding anode and cathode reactions in electrolysis. • Application of periodic table trends. • Writing observations for practical chemistry questions. • Identifying correct methods for salt preparation. • Applying Gay Lussac's Law, mole concept, and weight-volume relationships in numericals. • Confusion about the nature of oxides formed in a reaction. • Differentiating between reducing and oxidizing agents. • Identification of the gases evolved during the reaction. • Products formed in displacement, neutralization, or double displacement reactions. • Writing complete and accurate reasoning for observations or application-based questions. • Confusion in preferential discharge of ions during electrolysis. • Misinterpretation of diagrams or reactions due to incorrect notations in electron dot structures.

Suggestions for candidates

- Understand concepts thoroughly instead of merely memorizing facts.
- Focus on fundamentals -valencies, balanced equations, and correct formulae.
- Clarify doubts and misconceptions promptly to strengthen your conceptual base.
- Learn definitions with key scientific terms and precise wording.
- Create a realistic study schedule including weaker topics and regular revision.
- Study from the textbook; remember definitions and observations while practising.
- Practise a wide variety of numerical problems from all chapters.
- Solve sample and application-based questions regularly.
- Apply concepts to new situations to build confidence.
- Take timed mock tests to improve time management and accuracy.
- Practise MCQs and past papers to reinforce knowledge.
- Attempt competency-focused and observation-based questions uploaded on the CISCE website.
- Revise previous years' board and specimen papers for pattern familiarity.
- Create a reference chart of gas tests, observations, and their chemical equations.
- Learn all chemical reactions along with their conditions.
- Regularly practise writing and balancing equations correctly.
- Always mention reaction conditions when writing equations.
- Draw clear and accurate structural formulae for organic compounds.
- Revise organic chemistry — structures, reactions, and nomenclature thoroughly.
- Draw structures neatly and label clearly.
- Read every question carefully; identify key words before answering.
- Respond exactly to what is asked — avoid repetition or irrelevant details.
- Use correct terminology and provide precise, clear explanations.
- Focus on accuracy and clarity in application-based answers.
- Write neatly and legibly to ensure easy evaluation.
- Use reading time effectively to plan and prioritise answers.
- Recheck your paper after completion to avoid silly mistakes.
- Practise diagram-based questions with proper labelling.
- Check calculations carefully for correct multiplication and division.